

Electric Current

Notes

Electric Current :

- The rate of flow of electric charge in a conductor is termed as electric current.
- The direction of the flow of positive charge is considered the direction of flow of electric current.
- Thus, the direction of flow of electric current is just opposite to that the direction of flow of electrons.
- SI unit of electric current is Ampere.
1 Ampere = 1 coulomb/second.

Types of Electric Current :

- It is of two types-
 - (i) Direct current
 - (ii) Alternating current
- (i) **Direct Current (DC)** –
 - An electric current flowing in one direction only.
 - A battery is a suitable example of DC power supply.
- (ii) **Alternating Current (AC)** –
 - An electric current that reverses its direction many times a second at regular intervals, typically used in power supply.
 - Main advantage of AC than DC is that AC can be transmitted to a long distance without much electric energy loss.

Electromotive Force (emf) :

- Electromotive force is the electrical intensity or pressure developed by a source of electrical energy such as a battery or generator.

Voltage (Electric Potential) :

- The work done to bring a unit positive charge from an position to any point in the electric field is termed as the voltage of that particular point.
- Its SI Unit is Volt.

Potential Difference :

- The difference in electric potential between two points in an electric field; the work has to be done in transferring unit positive charge from one point to other.
- Its SI unit is also Volt.
- If work W joule is done to flow q charge between two points, then the potential difference between these points is-

$$V = \frac{\text{work done}}{\text{flow of charge}}$$

$$= \frac{W \text{ joule}}{q \text{ coulomb}}$$

$$= \frac{W}{q} \text{ joule / coulomb} = \frac{W}{q} \text{ volt}$$

Resistance :

- Resistance is a physical quality of conductor which resists the flow of electric current in the conductor.
- Its unit is Ohm (Ω).
- It is represented by 'R'.

Ohm's Law :

- Ohm's Law states that the current through a conductor between two points is directly proportional to the voltage across the two points.
- If V potential difference is applied at both ends of conductor and flow of electric current is I then,

$$I \propto V \text{ or } V \propto I$$

$$\Rightarrow V = RI$$

Where R is a constant, known as the resistance of conductor.

Factors Affecting the Resistance :

- (i) By experiments, it has been observed that resistance of conductor (R) is directly proportional to its length.

$$\text{or } R \propto L \quad \text{----- (i)}$$

- (ii) The resistance of conductor (R) is inversely proportional to its cross-sectional area (A).

$$\text{or } R \propto \frac{1}{A} \quad \text{----- (ii)}$$

Combining (i) & (ii) equations

$$R \propto \frac{L}{A}$$

$$R = p \frac{L}{A}$$

Where p is a proportional constant which is known as **Specific Resistance** or resistivity.

- Resistivity p of any conductor depends only upon the nature of matter.
- Its SI unit is Ohm-meter.

Electric Power :

- It is defined as the rate of consumption of electric energy in an electrical circuit.

$$\text{Electric Power} = \text{Electric current} \times \text{Potential difference}$$

$$\square P = VI$$

- The SI unit of electric power is Watt.

$$\Rightarrow 1 \text{ watt} = \frac{1 \text{ joule}}{1 \text{ coulomb}} \times \frac{1 \text{ coulomb}}{1 \text{ second}} = \frac{1 \text{ joule}}{1 \text{ second}}$$

$$\text{Electric Power} = \frac{\text{consumed electrical energy}}{\text{time}}$$

$$\square \text{ Electric energy} = \text{Power} \times \text{time}$$

- If we calculate electric power in watt and time in hour, then the unit of electrical energy will be, watt-hour.

- For commercial purposes–kilowatt-hour– a big unit of electrical energy is used.
- This is known as Unit.
- In other words-
"1 Unit electricity is the amount of electrical energy consumed by a load of 1 kilowatt power rating in 1 hour".

$$\text{No. of units} = \frac{\text{volt} \times \text{Ampere} \times \text{hour}}{1000} = \frac{\text{watt} \times \text{hour}}{1000}$$

Electric Generator :

- Electric Generator is a device which converts mechanical energy into electrical energy.
- Its application depends upon electromagnetic induction.

Transformer :

- It is a device based on electromagnetic induction.
- It increases or decreases the voltage of alternating current.
- Transformers are of two types :
(i) Step-up Transformer– it converts low voltage to high voltage of alternating current.
(ii) Step-down Transformer– it converts the high voltage into the low voltage of alternating current.
- Transformers are used for such electric devices which functions on voltage differing from main's voltage.

Electric Motor :

- An electric motor is a device that converts electrical energy into mechanical energy, usually by employing electromagnetic phenomena.
- Electric motors are used in electric fans, washing machines, mixer, grinder, water pumps and many other devices.
- Electric motors can be powered by DC sources, such as from batteries, motor vehicles or rectifiers, or by AC sources, such as power grid, inverters or electric generators.

Dynamo :

- A dynamo is an electrical generator that creates direct current using a commutator.
- It converts mechanical energy into electrical energy through electromagnetic induction.

Question Bank

1. Current density is :

- (a) a scalar quantity (b) a vector quantity
(c) dimensionless (d) None of the above

69th B.P.S.C. (Pre) 2023

Ans. (b)

The amount of electric current travelling per unit cross-section area is called as current density and expressed in amperes per square meter. It is a vector quantity. The current density vector is defined as a vector whose magnitude is the electric current per cross-sectional area at a given point in space, its direction being that of the motion of the positive charges at this point.

2. The dimension of dielectric constant is :

- (a) $M^0L^0T^0$ (b) ML^0T^0
(c) MLT^0 (d) M^0L^0T

Chhattisgarh P.C.S. (Pre) 2024

Ans. (a)

Dielectric constant or relative permittivity is the permittivity of a material expressed as a ratio with the electrical permittivity of a vacuum. The dielectric constant of a material is a measure of its ability to store electrical energy. It is a dimensionless quantity, hence option (a) is the correct answer.

3. An electron of mass M kg and charge e coulomb travels from rest through a potential difference of V volts. The final energy in joules would be :

- (a) eV (b) $\frac{eV}{M}$
(c) $\frac{e}{V}$ (d) MeV

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (a)

A particle with electric charge q passing through a potential difference (voltage) V gains an energy

$$E = q \cdot V$$

Here, q = e coulomb and V = V volts

Hence, final energy of the electron

$$E = eV \text{ coulomb-volt}$$

$$= eV \text{ joules}$$

if V = 1 volt and e = 1.602×10^{-19} coulomb

(Charge of an electron)

$$\text{Then } E = 1 \times 1.602 \times 10^{-19} \text{ coulomb-volt}$$

$$= 1.602 \times 10^{-19} \text{ joules}$$

4. A 100 watt electric bulb is used for 10 hours. What will be the cost of electricity consumed, if the consumption cost is Rs. 5 per unit ?

- (a) Rs. 5 (b) Rs. 10
(c) Rs. 25 (d) Rs. 50

R.A.S./R.T.S.(Pre) 2012

Ans. (a)

Energy Consumption of 100 watt electric bulb which is used for 10 hours = 100×10

$$= 1000 \text{ watt hour}$$

$$= 1 \text{ kilowatt hour}$$

$$= 1 \text{ unit.}$$

According to question the cost of 1 unit of electricity = Rs 5.

5. If a bulb of 100 watt burns for 10 hours, the expenditure of electricity will be –

- (a) 0-1 unit (b) 1 unit

(c) 10 units

(d) 100 units

41st B.P.S.C. (Pre) 1996

Ans. (b)

See the explanation of above question.

6. 15 bulbs each of 200 watt capacity in a Silkyara tunnel are illuminated continuously for 24 hours. The consumption of electricity will be :

(a) 71 units

(b) 72 units

(c) 73 units

(d) None of the above

U.P. R.O./A.R.O. (Pre) 2023

Ans. (b)

Consumption of electricity by one bulb

$$\begin{aligned} &= \frac{\text{watt} \times \text{hours}}{1000} \\ &= \frac{200 \times 24}{1000} \\ &= 4.8 \text{ units} \end{aligned}$$

Hence, total consumption of electricity by 15 bulbs

$$= 15 \times 4.8 = 72 \text{ units}$$

7. Five bulbs each of 100 watt capacity in a tunnel are illuminated continuously for twenty hours. The consumption of electricity will be:

(a) One unit

(b) Two units

(c) Ten units

(d) Twenty units

U.P.P.C.S. (Pre) 2017

Ans. (c)

Consumption of electricity by one bulb

$$\begin{aligned} (\text{No. of units}) &= \frac{\text{watt} \times \text{hours}}{1000} \\ &= \frac{100 \times 20}{1000} \\ &= 2 \text{ units} \end{aligned}$$

Thus, consumption of electricity by 5 bulbs = 2×5

$$= 10 \text{ units}$$

8. The value of 1 kilowatt hour is –

(a) 3.6×10^6 J

(b) 3.6×10^3 J

(c) 10^3 J

(d) 10^5 J

U.P.P.C.S. (Pre) 2006

Ans. (a)

1 Kilowatt hour = 1 (kilowatt) (1 hour)

= (1000 joule/sec.) (3600 sec.)

= 36×10^5 Joule

= 3.6×10^6 Joule

Thus, option (a) is correct.

9. An electric bulb of 100 watt is used for 4 hours. The unit of electric energy used is –

(a) 400

(b) 25

(c) 4

(d) 0.4

U.P. Lower Sub. (Pre) 2009

Ans. (d)

$$\begin{aligned} \text{The electric energy consumed} &= 100 \text{ watt} \times 4 \text{ hour} \\ &= 400 \text{ watt hour} \\ &= \frac{400}{1000} \text{ kilowatt hour} \\ &= 0.4 \text{ kilowatt hour} \\ &= 0.4 \text{ unit} \end{aligned}$$

10. Electricity consumption bill is based on the measurement of –

1. Watt

2. Voltage

3. Ohm

4. Ampere

Select your answer of the following codes :

(a) Only 1

(b) 1 and 2

(c) 2 and 3

(d) 1 and 4

40th B.P.S.C. (Pre) 1995

Ans. (a)

Electricity consumption bill is based on the measurement of kilowatt hour. The kilowatt hour is the unit of electricity consumption, equal to 1000 watt in 1 hour (1 unit).

11. How many units of electricity will be consumed if you use a 60-watt electric bulb for 5 hours everyday for 30 days?

(a) 12

(b) 9

(c) 6

(d) 3

R.A.S./R.T.S. (Pre) 2003

Ans. (b)

We know that –

Number of units = watt $\times$ hour / 1000

$$= \frac{60 \times 30 \times 5}{1000} = 9 \text{ unit}$$

12. An electric bulb is connected to 220 v generator. The current is 0.5 A. The power of the bulb is –

(a) 110 w

(b) 110 v

(c) 440 w

(d) 220 w

(e) None of these

Chhattisgarh P.C.S. (Pre) 2016

Ans. (a)

Given,

$$V = 220 \text{ V}$$

$$I = 0.5 \text{ A}$$

we know that,

$$P = \frac{W}{t} = \frac{VIt}{t} \quad (\because W = VIt)$$

$$= VI = 220 \times 0.5 \Rightarrow 110 \text{ watt}$$

13. Which of the following is the resistance of the wire?

(a) $R = IV$

(b) $R = I/2V$

(c) $R = V/I$

(d) $R = I/V$

70th B.P.S.C. (Pre) 2024

Ans. (c)

The resistance (R) of a wire is calculated using Ohm's Law :

$R = V/I$, where V is the voltage (potential difference) across the wire and I is the current flowing through it.

Ohm's Law states that the current through a conductor between two points is directly proportional to the voltage across the two points. When the physical conditions (like temperature) are constant, the ratio of voltage to current, or V/I , is a constant value, which we call resistance (R).

14. Of the two bulbs in a house, one glows brighter than the other. In this context which of the following statements is correct ?

(a) The brightness does not depend on the resistance.

(b) Both the bulbs have same the resistance.

(c) The bright bulb has larger resistance.

(d) The dim bulb has larger resistance.

U.P. Lower Sub. (Pre) 2008

U.P.P.C.S. (Pre) 2009

Ans. (d)

The bulb which decays more energy will generate more brightness.

Power (P) = Voltage (V) $\times$ Current (I)

It is known that $V = I \times \text{Resistance (R)}$

$$P = \frac{V^2}{R}; P \propto \frac{1}{R}$$

Thus, the bulb of high resistance decays low energy and will generate dim light.

15. Of the two bulbs in a house, one glows brighter than the other. Which of the two has a larger resistance ?

(a) The dim bulb

(b) The brighter bulb

(c) Both have the same resistance

(d) The brightness does not depend on the resistance

U.P.P.C.S. (Mains) 2013

Ans. (a)

See the explanation of above question.

16. How two identical bulbs should be connected to get maximum light?

1. Both are connected in series.

2. Both are connected in parallel.

Select the correct answer using the code given below :

Code :

(a) In case 1 more

(b) In case 2 more

(c) Same in both 1 and 2

(d) Nothing can be said

U.P. R.O./A.R.O. (Pre) 2021

Ans. (b)

Bulbs in parallel are brighter than bulbs in series. In a parallel circuit the voltage for each bulb is the same as the voltage in the circuit while the parallel combination of resistors reduces the effective resistance of the circuit. The power dissipated by a bulb is given by

$$P = V^2/R,$$

where V is the voltage of the light bulb and R is the electrical resistance.

The total resistance of the series circuit is

$$R_{\text{series}} = R + R = 2R.$$

The total resistance of the parallel circuit is

$$1/R_{\text{parallel}} = 1/R + 1/R$$

or

$$R_{\text{parallel}} = R/2.$$

Thereby, the power for the light bulbs in series is

$$P_{\text{series}} = V^2/2R$$

and the power for the light bulbs in parallel is

$$P_{\text{parallel}} = 2V^2/R$$

Hence, from the above equations it is clear that in parallel circuit the bulbs will shine brightly.

17. The total resistance of a circuit having two parallel resistors is 1.403 kilo-ohm. If one of the resistors is 2.0 kilo-ohm, then the other resistor will be

(a) 1.403 kilo-ohm

(b) 2.0 kilo-ohm

(c) 3.403 kilo-ohm

(d) 4.70 kilo-ohm

(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (d)

If total resistance of a parallel circuit is R, then

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

According to the question,

$$\frac{1}{1.403} = \frac{1}{2} + \frac{1}{R_2}$$

$$\text{or, } \frac{1}{R_2} = \frac{1}{1.403} - \frac{1}{2}$$

$$\text{or, } \frac{1}{R_2} = \frac{2 - 1.403}{2.806}$$

$$\text{or, } R_2 = \frac{2.806}{0.597} = 4.70 \text{ kilo-ohm}$$

18. A voltage of 15 volt is applied across a colour coded carbon resistor with first, second and third rings of brown, green and black colours. What is the current flowing through the resistor?

- (a) 3 ampere (b) 2 ampere
(c) 1 ampere (d) None of the above

U.P. R.O./A.R.O. (Pre) 2023

Ans. (c)

Resistor code value for brown colour = 1 (A)

Resistor code value for green colour = 5 (B)

Resistor code value for black colour = 0 (C)

We know that first two bands (rings) indicate the first two significant digits of resistance value and the third band serves as a multiplier.

$$\begin{aligned} \text{So, Resistance value of resistor} &= (A \times 10 + B) \times 10^C \\ &= (1 \times 10 + 5) \times 10^0 \\ &= 15 \times 1 = 15 \text{ Ohm } (\Omega) \end{aligned}$$

$$\text{Current (i)} = \frac{\text{Voltage (V)}}{\text{Resistance (R)}}$$

$$= \frac{15}{15} = 1 \text{ ampere}$$

Hence, the current flowing through the resistor is 1 ampere.

19. Reactance of a capacitor is :

- (a) Directly proportional to frequency
(b) Inversely proportional to sq. root of frequency
(c) Inversely proportional to frequency
(d) None of the above

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (c)

The reactance of a capacitor (Capacitive reactance – X_c) is inversely proportional to the frequency.

$$X_c = \frac{1}{\omega C} \quad (\text{where } \omega = \text{Angular frequency, } C = \text{Capacitance})$$

$$X_c = \frac{1}{2\pi fC} \quad (\text{where, } \omega = 2\pi f)$$

$$\square X_c \propto \frac{1}{f}$$

The reactance of an inductor (X_L) is directly proportional to the frequency.

$$X_L = \omega L = 2\pi fL \quad (\text{where } L = \text{Inductance})$$

$$\square X_L \propto f$$

20. A dynamo which is said to generate electricity actually acts as a :

- (a) Source of ions (b) Source of electric charge
(c) Converter of energy (d) Source of electrons

U.P.P.C.S. (Mains) 2009

Ans. (c)

A dynamo, which is also known as an electrical generator produces direct current through a commutator. It is basically a device which converts mechanical rotation into electric current according to Faraday's law.

21. 'Dynamo' is a device, which converts :

- (a) Chemical energy into electrical energy
(b) Electrical energy into mechanical energy
(c) Mechanical energy into electrical energy
(d) Electrical energy into chemical energy

U.P. P.C.S. (Pre) 2016

39th B.P.S.C. (Pre) 1994

U.P.P.C.S. (Pre) 1993

U.P.P.C.S. (Pre) 1992

Ans. (c)

A dynamo is an electrical generator that produces direct current with the use of a commutator. It converts mechanical energy into electrical energy.

22. Dynamo is a machine which is used for –

- (a) Conversion of high voltage to low voltage
(b) Conversion of electrical energy to mechanical energy
(c) Conversion of mechanical energy to electrical energy
(d) Conversion of low voltage to high voltage

U.P.P.C.S. (Mains) 2012

R.A.S./R.T.S.(Pre) 2012

Ans. (c)

See the explanation of above question.

23. Turbines and Dynamos are used to convert which energy to electrical energy?

- (a) Chemical energy (b) Solar energy
(c) Mechanical energy (d) Magnetic energy

44th B.P.S.C. (Pre) 2000

Ans. (c)

See the explanation of above question.

24. When electrical energy is converted into motion

- (a) There is no heat loss
- (b) Heat loss is 50 percent
- (c) Heat loss is 30 percent
- (d) Heat loss is 80 percent

47th B.P.S.C. (Pre) 2005

Ans. (a)

When electrical energy is converted into motion, there is no heat loss.

25. A device which converts electrical energy into mechanical energy is –

- (a) Dynamo
- (b) Transformer
- (c) Electric motor
- (d) Inductor

U.P.P.C.S. (Mains) 2007

Uttarakhand P.C.S. (Pre) 2005

Ans. (c)

Electric motor is a device which converts electrical energy into mechanical energy. Electric motors involve rotating coils of wire which are driven by the magnetic force exerted by a magnetic field or an electric current.

26. In electrical motor :

- (a) heat is converted into electrical energy
- (b) electrical energy is converted into heat
- (c) electrical energy is converted into mechanical energy
- (d) mechanical energy is converted into electrical energy
- (e) None of the above / More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (c)

See the explanation of above question.

27. The device which works on the principle of electromagnetic induction is

- (a) Ammeter
- (b) Voltmeter
- (c) Dynamo
- (d) Electric motor

U.P. R.O./A.R.O. (Pre) 2017

Ans. (c) & (d)

Among the above options, both Dynamo and Electric motor are such devices that work on the principle of electromagnetic induction. Dynamo converts mechanical energy into electrical energy and electric motor converts electrical energy into mechanical energy.

28. A device which converts chemical energy into electrical energy is called –

- (a) Battery
- (b) Motor

(c) Generator

(d) Moving coil meter

R.A.S./R.T.S. (Pre) 2013

Ans. (a)

An electrical battery is a device consisting of two or more electrochemical cell that converts stored chemical energy into electrical energy. There are different types of batteries but all have three basic components; a positive electrode, negative electrode, and electrolyte.

29. Electric motors operating at low voltages tend to burn out because –

- (a) They draw more current which is inversely proportional to the voltage.
- (b) They draw more current which is inversely proportional to the square root of the voltage.
- (c) They draw heat proportional to V^2 .
- (d) Low voltage sets in electrical discharge.

U.P.P.C.S. (Pre) 2008

Ans. (a)

Electric motors operating at low voltages tend to burn out because they draw more current which is inversely proportional to the voltage.

30. Faraday constant :

- (a) depends on the amount of the electrolyte
- (b) depends on the current passed in the electrolyte
- (c) depends on the volume of the solvent in which the electrolyte is dissolved
- (d) is a universal constant
- (e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (d)

The Faraday constant represents the magnitude of electric charge per mole of electrons. It is denoted by the symbol 'F' and is named after English scientist Michael Faraday. It is a universal constant. It has the currently accepted value $F = 96485.332123 \text{ C.mol}^{-1}$.

31. Consider the following statements regarding a motor car battery -

1. The voltage is usually 12 V.
2. Electrolyte used is hydrochloric acid.
3. Electrodes are lead and copper.
4. Capacity is expressed in ampere-hour.

Which of the above statements are correct ?

- (a) 1 and 2
- (b) 2 and 3
- (c) 3 and 4
- (d) 1 and 4

I.A.S. (Pre) 1999

Ans. (d)

A motor car battery usually has six 2V cells connected in series, which gives it a steady voltage of about 12V. The capacity of battery is expressed in ampere-hour. An ampere-hour is the total amount of electrical charge transferred when a current of one ampere flows for one hour. Therefore the total usable charge stored in a battery can be stated in term of ampere-hour. The electrolyte used is a approx. 35% solution by weight of sulphuric acid H_2SO_4 with 65% of water. The electrodes used in motor car batteries are lead (Pb) and lead dioxide (PbO_2). Hence, only statements 1 and 4 are correct.

32. Nickel-Cadmium (Ni-Cd) battery is used in :

- (a) Calculator
- (b) Cordless electronic appliances
- (c) Transistors
- (d) All of the above

Uttarakhand P.C.S. (Pre) 2016

Ans. (d)

The nickel-cadmium battery is a type of rechargeable battery using nickel oxide hydroxide and metallic cadmium as electrodes, while potassium hydroxide is used as alkaline electrolyte. It is used in calculator, cordless electronic appliances, transistors, portable power tools, photography equipments, flashlight etc.

33. Which of the following metals is used as filament in lighting bulbs?

- (a) Iron
- (b) Molybdenum
- (c) Silver
- (d) Tungsten

44th B.P.S.C. (Pre) 2000

Ans. (d)

Tungsten metal is used as filament in lighting bulbs. In 1781, Carl Wilhelm Scheele and Torbern Bergman suggested that it might be possible to obtain a new metal by reducing tungsten acid. Two years later in 1783, Juan Jose and Fausto Elhuyar were the first to isolate tungsten through the reduction of this acid with charcoal. For this reason, they are credited with the discovery of this element. It's melting point is $3422^\circ C$ and it's boiling point is $5555^\circ C$.

34. The filament of an electric bulb is made of –

- (a) Magnesium
- (b) Iron
- (c) Nichrome
- (d) Tungsten

U.P.P.C.S (Pre) 2011

U.P.P.C.S. (Pre) 2005

U.P.P.C.S. (Pre) 1990

Ans. (d)

See the explanation of above question.

35. Filament of electric bulb is made of :

- (a) Tungsten
- (b) Nichrome
- (c) Lead
- (d) Aluminium

Uttarakhand P.C.S. (Pre) 2012

Ans. (a)

See the explanation of above question.

36. Consider the following statements :

An ordinary light bulb has a rather short life because the:

1. Filament wire is not uniform.
2. Bulb cannot be evacuated completely.
3. Wires supporting the filament melt at high temperatures.

Which of the above statements are correct ?

- (a) 1 and 3
- (b) 2 and 3
- (c) 1 and 2
- (d) 1, 2 and 3

I.A.S. (Pre) 1999

Ans. (d)

An ordinary light bulb has a rather short life because the (1) filament wire is not uniform, (2) bulb cannot be evacuated completely and (3) wires supporting the filament melt at high temperature.

37. The temperature of the filament of lighted electric bulb is generally :

- (a) $100^\circ C$ to $500^\circ C$
- (b) $1000^\circ C$ to $1500^\circ C$
- (c) $2000^\circ C$ to $2500^\circ C$
- (d) $3000^\circ C$ to $3500^\circ C$

R.A.S./R.T.S.(Pre) 2007

Ans. (c)

Tungsten metal is used as filament in electric bulb. When electric current flows through it, its temperature goes up to $1500^\circ C$ to $2500^\circ C$. Generally, electric bulbs convert only 5% to 10% of electric energy into light.

38. Lightning can even burn a tree because it contains tremendous amount of

- (a) Heat energy
- (b) Electric energy
- (c) Chemical energy
- (d) Nuclear energy

M.P.P.C.S. (Pre) 2017

Ans. (b)

Lightning can even burn a tree because it contains tremendous amount of electrical energy. Lightning is an electrical discharge caused by imbalance between storm clouds and the ground, or within the clouds themselves.

39. Consider the following statements :

Assertion (A) : Lightning conductors prevent buildings from damages in the event of lightning strike.

Reason (R) : The electric charge conduct to ground through the wire, instead of passing through the structure.

Select the correct answer using the codes given below:

Code :

- (a) Both (A) and (R) are correct, and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are correct, but (R) is not the correct explanation of (A).
- (c) (A) is correct but (R) is wrong.
- (d) (A) is correct but (R) is wrong.

Uttarakhand P.C.S.(Mains) 2002

U.P.P.C.S. (Pre) 1999

Ans. (a)

A lightning conductor is a metal rod mounted on the top of an elevated structure, electrically bonded using a wire or electrical conductor to interface with ground through an electrode, engineered to protect the structure in the event of lightning strike. If lightning hits the structure, it will preferentially strike the rod and would be conducted to ground through wire instead of passing through the structure. Hence the building will be saved by using this lightning conductor.

40. In a three-pin electrical plug longest pin should be connected to

- (a) Ground terminal
- (b) Live terminal
- (c) Neutral terminal
- (d) Any terminal

U.P.P.C.S. (Pre) 2007

Ans. (a)

In the three-pin plug, there are three ends namely live, neutral and earth. The earth pin on a plug is longer than the line and neutral pins. This means the earth pin is the first to connect and last to disconnect for safety of electrical device. It provides an alternative and easy path for leakage or fault current flow.

41. In electric apparatus 'earth' is used:

- (a) To reduce the expenditure
- (b) Because the apparatus used 3-phase
- (c) For safety
- (d) In the form of fuse

U.P.P.C.S. (Pre) 2002

Ans. (c)

See the explanation of above question.

42. The electric current does not flow between two properly connected charged bodies if they are having

- (a) Same charge
- (b) Same capacity
- (c) Same resistivity
- (d) Same potential

U.P.P.S.C. (R.I.) 2014

Ans. (d)

According to Ohm's law, if there is a potential difference (V) across a resistor then there is a current (I) flowing through it. Current flows in a circuit as a result of difference in potential between two points in the circuit.

43. A fuse is used in main electric supply as a safety device. Which one of the following statements about the fuse is correct?

- (a) It is connected in parallel with the main switch
- (b) It is made mainly from silver alloys
- (c) It must have a low melting point
- (d) It must have a very high resistance

I.A.S. (Pre) 1998

Ans. (c)

An electric fuse is a safety device used to prevent damage to an electrical circuit when excessive current flows through it. The electric fuse works on the principle of the heating effect of electric current. It consists of a piece of wire made of a metal or an alloy of low melting point. As the current increases beyond a limit, the wire in the electric fuse melts and breaks off. The fuse is then said to have blown off. The circuit is broken and current stop flowing through it. Thus, a fuse prevents fires.

44. Principle of fuse is

- (a) Chemical effect of electricity
- (b) Mechanical effect of electricity
- (c) Heating effect of electricity
- (d) Magnetic effect of electricity

43rd B.P.S.C. (Pre) 1999

Ans. (c)

See the explanation of above question.

45. A fuse wire is characterized by:

- (a) Low resistance and high melting point.
- (b) Low resistance and low melting point.
- (c) High resistance and low melting point.
- (d) High resistance and high melting point.

U.P. P.C.S. (Pre) 1993

Ans. (c)

Fuse is a piece of wire with high resistance and very low melting point. When a high current flows through the circuit, it gets heated and melts.

46. Fuse wire used for safety of household have :

- (a) Low melting point (b) High melting point
- (c) High resistance (d) Low resistance.

R.A.S./R.T.S. (Pre) 1997-98

Ans. (a) & (c)

See the explanation of above question.

47. The filament of a halogen lamp is an alloy of –

- (a) Tungsten and iodine
- (b) Tungsten and bromine
- (c) Tungsten and sodium
- (d) Molybdenum and sodium

R.A.S./R.T.S.(Pre) 1999

Ans. (c)

The filament of a halogen lamp is an alloy of tungsten and sodium. Due to the presence of sodium in filament, it produces yellow light.

48. Assertion (A) : The temperature of a metal wire rises when an electric current is passed through it.

Reason (R) : Collision of metal atoms with each other releases heat energy.

Code :

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (b) Both (A) and (R) are true but (R) is not a correct explanation of (A)
- (c) (A) is true but (R) is false
- (d) (A) is false but (R) is true

I.A.S. (Pre) 1998

Ans. (c)

When electric current passes through the conducting (metal) wire, moving electrons continuously collide with the atoms of the conductor. In this process, they transfer their energy to the atoms of the conductor and therefore, temperature of the conductor increases. Thus, assertion (A) is correct but reason (R) is false.

49. Domestic electrical wiring is basically a :

- (a) Series connection
- (b) Parallel connection
- (c) Combination of series and parallel connections
- (d) Series connection within each room and parallel connection elsewhere

I.A.S. (Pre) 1996

Ans. (b)

Domestic electrical wiring is basically a parallel connection. In parallel circuit, each branch receives equal current and resistance is minimum. In a parallel connection if one of the bulbs is removed from the circuit i.e. if one of the branches in the circuit is broken, electric current will still flow in other branches and other bulbs will continue to glow.

50. A wire with black insulation during electric supply is–

- (a) Live wire (b) Earth wire
- (c) Neutral wire (d) Fuse wire
- (e) None of the above

Chhattisgarh.P.C.S. (Pre) 2014

Ans. (c)

The electric power line reaches our house through three wires namely a live wire, neutral wire, and earth wire. To avoid confusion we follow a colour code for insulating these wires. The red wire is a live wire, and the black wire is neutral. The earth wire is given green plastic insulation.

51. What is the order of magnitude of electric resistance of the human body (dry) ?

- (a) 10^2 ohm (b) 10^4 ohm
- (c) 10^6 ohm (d) 10^8 ohm

I.A.S. (Pre) 2005

Ans. (b)

The order of magnitude of electric resistance of the human body (dry) is generally in the range of 10^4 (10000) ohms and it may be as high as 10^5 (100000) ohms or more. Wet or broken skin may drop the body's resistance to 1000 ohms or less.

52. Full form of C.F.L. is –

- (a) Compact Fluorescent Lamp
- (b) Centrally Fixed Lamp
- (c) Chemical Fluorescent Lamp
- (d) Condensed Fluorescent Lamp

U.P.P.C.S. (Mains) 2015

Ans. (a)

C.F.L. is short form of Compact Fluorescent Lamp. CFL uses significantly less energy than traditional light bulbs (about 75% less).

53. Light Emitting Diode (LED) work on principle of :

- (a) Laser (b) Thermionic emission
- (c) Photoelectric code (d) Electroluminescence

Jharkhand P.C.S. (Pre) 2021

Ans. (d)

Light Emitting Diode (LED) work on the Principle of Electroluminescence. Electroluminescence (EL) is an optical and electrical phenomenon in which a material emits light in response to the passage of an electric current or to a strong electric field. It is the result of radiative recombination of electrons and holes in a material, usually a semiconductor. The excited electrons release their energy as photons – light. A light-emitting diode (LED) is a semiconductor light source that emits light when current flows through it. Electrons in the semiconductor recombine with electron holes, releasing energy in the form of photons.

54. Assertion (A): Light Emitting Diode (LED) lamps offer longer service life as compared to Compact Fluorescent Lamps (CFL).

Reason (R) : Light Emitting Diodes (LED) lamps are more energy efficient as compared to the Compact Fluorescent Lamps (CFL).

Select the correct answer using the codes given below:

- (a) Both (A) and (R) are true, and (R) explains (A).
- (b) Both (A) and (R) are true, but (R) does not explain (A).
- (c) (A) is true, and (R) is false.
- (d) (A) is false, and (R) is true.

R.A.S./R.T.S.(Pre) 2013

Ans. (a)

In Light Emitting Diode (LED) Lamps, gallium arsenide is used as main light emission component which converts electric energy into light. Its life span is greater than that of CFL because LED lamps consume less energy than CFL.

55. What is the difference between a CFL and an LED Lamp?

- 1. To produce light, a CFL uses mercury vapour and phosphor while an LED lamp uses semi-conductor material.
- 2. The average life span of a CFL is much longer than of an LED lamp.
- 3. A CFL is less energy-efficient as compared to an LED lamp.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 and 3
- (c) 1 and 3
- (d) 1, 2 and 3

I.A.S. (Pre) 2011

Ans. (c)

CFL uses mercury vapour and phosphor to produce light. The gases become excited by electricity and when combined with mercury particles, produce invisible ultraviolet light. The UV light then hits the white phosphor coating inside the bulb causing it to fluoresce and emit white light. LED lamp uses semi-conductor material to produce light. CFLs have a life of about 6000 to 15,000 hours, while LED has a life span of about 50,000 to 100,000 hours or more. LED lamps are more energy efficient as compared to CFL.

56. With reference to street-lighting, how do sodium lamps differ from LED lamps?

- 1. Sodium lamps produce light in 360 degrees but it is not so in the case of LED lamps.
- 2. As street-lights, sodium lamps have longer life span than LED lamps.
- 3. The spectrum of visible light from sodium lamps is almost monochromatic while LED lamps offer significant colour advantages in street-lighting.

Select the correct answer using the code given below.

- (a) 3 only
- (b) 2 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

I.A.S. (Pre) 2021

Ans. (c)

Lights from sodium lamps i.e. sodium vapour lights are omnidirectional. Omnidirectional lights produce light in 360 degrees. This is a large system inefficiency because at least half of the light needs to be reflected and redirected to the desired area being illuminated. Whereas LED lamps emit light directionally i.e. over 180 degrees which means there are far fewer losses from the need to redirect or reflect light. Hence, statement 1 is correct.

LEDs have an extremely long lifespan relative to every other lighting technology (including sodium lamps). New LEDs can last about 50,000 to 100,000 hours or more i.e. more than 2-3 times than sodium lamps. Hence, statement 2 is incorrect. Sodium vapour lights have the worst colour rendering. Low pressure sodium lamps in particular are monochromatic resulting in the colours of illuminated objects being nearly indistinguishable, which means that the objects illuminated by them appear shadowy black rather than the colour you would see during daylight. LEDs waste much less energy producing waste heat and they also provide an incredibly better variety of high colour rendering index options to the user (thus eliminating the monochromatic black appearance of objects illuminated by sodium lamps). Hence, statement 3 is correct.

57. What gases are filled in fluorescent tube?

- (a) Neon (b) Sodium
(c) Mercury (d) Mercury and neon

U.P. P.C.S. (Pre) 1990

Ans. (d)

Fluorescent lamps are filled with the low pressure gases specially Mercury and noble gases like Argon, Neon, Xenon and Krypton.

58. In comparison to an electric bulb, a fluorescent tube is preferred because –

- (a) It has greater external outer surface to radiate/emit the light.
(b) There is no effect of voltage fluctuation on it.
(c) In the tube, electric energy almost converted into light energy.
(d) None of these.

U.P. P.C.S. (Pre) 2009

Ans. (d)

In comparison to electric bulb, fluorescent tubes transform more electric energy into light energy. A bulb of 100 watt converts only 5 percent of electric energy into light energy but fluorescent tubes convert approximately 22% of electric energy into light energy. Thus, none of the statement is true.

59. How does the white light produce in the tube?

- (a) by heating up the copper wire
(b) by heating up the filament
(c) by accelerating the atoms
(d) by oscillating the molecules

47th B.P.S.C. (Pre) 2005

Ans. (d)

A tube light or fluorescent lamp is a kind of light that uses a combination of electricity and mercury vapour to create light energy. The electricity causes electrons to travel at high speed between the poles. These electrons hit the mercury gas molecules and oscillate them. The mercury gas is then ionized by the current flowing within, producing ultraviolet radiation. Since the human eye is unable to see this radiation, the glass tube is coated with a fluorescent material which changes it into white light. Thus, among the given options, option (d) is the correct answer.

60. Which is the gas inside an electric bulb?

- (a) Oxygen (b) Air
(c) Carbon dioxide (d) None of the above

M.P.P.C.S. (Pre) 2000

Ans. (d)

The electric bulb was invented by Edison. A coiled filament of tungsten metal with a high melting point fits inside it. To prevent the metal from oxidation, Nitrogen and Argon gases are filled in the bulb instead of vacuum.

61. Fluorescent tubes are fitted with a choke. The choke coil –

- (a) Steps up the line voltage
(b) Steps down the line voltage
(c) Reduces current in the circuit
(d) Chokes low frequency current

I.A.S. (Pre) 2000

Ans. (a)

Starter in tube lights increases the current and sends it to the choke. Choke coil steps up the voltage of the transmitted current and send it back to tubelight resulting tubelight to glow. After this work of the starter gets finished and choke is used to control the voltage thus prevent the tubelight from any damage.

62. With reference to the electric potential which of the following statements is/are correct?

1. It is a scalar quantity.
2. It is a vector quantity.

Select the correct answer using the codes given below.

Codes :

- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

U.P. P.C.S. (Pre) 2021

Ans. (a)

Electric potential is a scalar quantity. It is the amount of work needed to move a unit charge from a reference point to a specific point against an electric field.

63. If potential difference and current flowing through a wire is increased twice each then electric power :

- (a) Will remain unaffected (b) Will increase four times
(c) Will increase two times (d) Will decrease half
(e) None of these

Chhattisgarh P.C.S. (Pre) 2015

Ans. (b)

Power $P = V \times I$
where $V =$ Voltage
 $I =$ Current
Since both the voltage and current are increased twice
 $P' = (2V)(2I)$
 $P' = 4 V \times I = 4P$
Hence, the electric power will increase four times.

64. Two wires have their lengths, diameters and resistivities all in the ratio of 1 : 2. If the resistance of the thinner wire is 10 ohm, the resistance of the thicker wire is :

- (a) 5 ohm (b) 10 ohm
(c) 20 ohm (d) 40 ohm

I.A.S. (Pre) 2001

Ans. (b)

We know that,

$$R = \rho \frac{l}{a}$$

R = resistance, ρ = resistivity, l = length, a = Area.

Let the ρ_1 , l_1 and r_1 is resistivity, length, and radius of first wire respectively then,

As per the question

Resistivity of second wire $\rho_2 = 2\rho_1$

Length of second wire $l_2 = 2l_1$

Radius of second wire $r_2 = 2r_1$

$$\Rightarrow R_1 = \frac{\rho_1 l_1}{a_1} = \frac{\rho_1 l_1}{\pi r_1^2}$$

$$R_2 = \frac{\rho_2 l_2}{a_2} = \frac{2\rho_1 \times 2l_1}{\pi (2r_1)^2}$$

$$= \frac{4\rho_1 l_1}{4\pi r_1^2} = \frac{\rho_1 l_1}{\pi r_1^2}$$

$$\Rightarrow \frac{R_1}{R_2} = 1$$

Since resistance of thin wire $R_1 = 10\Omega$ (ohm)

Therefore, resistance of thicker wire $R_2 = 10\Omega$ (ohm)

65. Small drops of the same size are charged to V volts each. If n such drops coalesce to form a single large drop, its potential will be :

- (a) $n^{2/3} V$ (b) $n^{1/3} V$
(c) $n V$ (d) $n^{-1} V$

U.P. Lower Sub. (Mains) 2015

Ans. (a)

Let the charge on every small drop = q

Radius of every small drop = r

Charge on large drop (Q') = nq

Radius of large drop = R

$\therefore$ the total volume of n drops = volume of the large drop

$$\therefore n \times \frac{4}{3} \pi r^3 = \frac{4}{3} \pi R^3$$

$$\square R = n^{1/3} \times r$$

$$\text{Now, } \frac{V'}{V} = \frac{Q'C}{qC} \quad (\text{where } C \text{ is Capacitance})$$

$$= \frac{Q'}{q} \times \frac{r/K}{R/K} \quad (\text{where } K \text{ is dielectric constant})$$

$$= \frac{nq}{q} \times \frac{r}{n^{1/3} \times r}$$

$$\square V' = n^{2/3} \times V$$

$$\square \text{ So, potential of large drop} = n^{2/3} V$$

66. Consider a tightly wound 100 turn coil of radius 10 cm, carrying a current of 1A. What will be the magnitude of the magnetic field at the centre of the coil?

- (a) $6.28 \times 10^{-4} \text{ T}$ (b) $6.28 \times 10^{-3} \text{ T}$

(c) $2.68 \times 10^{-4} \text{ T}$

(d) $6.82 \times 10^{-2} \text{ T}$

Uttarakhand P.C.S. (Pre) 2024

Ans. (a)

Given that, Number of turns $N = 100$

Radius of the coil $r = 10 \text{ cm} = 0.10 \text{ m}$

Current $I = 1 \text{ A}$

Magnetic field at the centre of the coil $B = \frac{\mu_0 NI}{2r}$

$$\therefore \mu_0 \text{ (Permeability of free space)} = 4\pi \times 10^{-7} \frac{\text{T.m}}{\text{A}}$$

$$\therefore B = \frac{(4\pi \times 10^{-7} \frac{\text{T.m}}{\text{A}})(100)(1\text{A})}{2(0.1\text{m})}$$

$$= \frac{4\pi \times 10^{-5} \text{ T.m}}{0.2 \text{ m}}$$

$$= 2\pi \times 10^{-4} \text{ T}$$

$$= 2 \times (3.14) \times 10^{-4} \text{ T (Taking } \pi = 3.14)$$

$$= 6.28 \times 10^{-4} \text{ T}$$

Hence, option (a) is the correct answer.

67. Given below are two statements, one labelled as Assertion (A) and the other labelled as Reason (R).

Assertion (A) : Transformer is useful for stepping up or stepping down voltages.

Reason (R) : Transformer is a device used in D.C. circuits.

In the context of the above two statements, which one of the following is correct ?

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is not a correct explanation of (A).
(c) (A) is true but (R) is false
(d) (A) is false but (R) is true

I.A.S. (Pre) 1996

Ans. (c)

A transformer transfers electric energy between two or more interconnected circuits through electric induction. It converts the high voltage to low voltage and low voltage to high and transmits it. Transformers can only function with alternating current (AC) not with direct current (DC).

68. Transformer is used for :

- (a) Converting AC into DC
(b) Converting DC into AC
(c) To step-up DC voltages
(d) To step-up or step-down AC voltages

U.P.P.C.S. (Pre) 2006

Ans. (d)

The transformer is used to step-up and step-down the alternating current. There are two types of coils in transformers (i) Primary coil (2) Secondary coil. Alternating current flows through the primary coil which inflicts the potential in the secondary coil.

69. Which of the following is the purpose for which a transformer is used ?

- (a) To convert AC into DC
- (b) To convert DC into AC
- (c) To step-up or step-down AC voltage
- (d) To step-up DC voltage

U.P.P.C.S. (Mains) 2011

Ans. (c)

See the explanation of above question.

70. A mobile phone charger is :

- (a) An inverter
- (b) A UPS
- (c) A step up transformer
- (d) A step down transformer

U.P.P.C.S. (Pre) 2017

Ans. (d)

A mobile phone charger works on the simple principle of conversion of AC (alternating current) to DC (direct current). As we connect charger to 220 volt AC switchboard, the first job of the charger is to step down the high 220 v in 9 v or 10v. A step down transformer is used in mobile phone charger.

71. Some pairs of devices and associated energy changes are given below :

- (i) Electric generator – Mechanical to electrical
- (ii) Electric motor – Electrical to mechanical
- (iii) Diesel engine – Light to electrical
- (iv) Solar cell – Chemical to mechanical

Which of the above pair/s is/are correctly matched?

- (a) (i) and (ii)
- (b) (i) and (iii)
- (c) (ii) and (iv)
- (d) (iii) and (iv)

R.A.S./R.T.S.(Pre) 2013

Ans. (a)

An electric generator is a device that converts mechanical energy obtained from an external source into electrical energy as the output. A motor converts electrical energy to mechanical energy. A diesel engine is an internal-combustion engine in which heat produced by the compression of air in the cylinder is used to ignite the fuel. Solar (or photovoltaic) cells convert the sun's energy into electricity.

72. An AC current can be produced by :

- (a) choke coil
- (b) dynamo

(c) transformer

(d) None of the above

69th B.P.S.C. (Pre) 2023

Ans. (b)

Alternating current (AC) is an electric current that periodically reverses direction and changes its magnitude continuously with time, in contrast to direct current (DC), which flows only in one direction. An AC current is produced by a dynamo which is called AC dynamo or alternator. This is done by rotating a coil or armature within the magnetic field created by field coils. If a DC current is required it can be supplied by a commutator in a DC dynamo. It is important to note that the current produced by a simple dynamo is alternating current (AC), as the direction of the current changes with each half turn of the coil. However, generally dynamos incorporate a component called a commutator, which converts the AC into direct current (DC).

73. The basic difference between an AC generator and a DC generator is that -

- (a) AC generator has an electromagnet while a DC generator has permanent magnet
- (b) AC generator has slip rings while the DC generator has a commutator
- (c) DC generator will generate a higher voltage
- (d) AC generator will generate a higher voltage

Chhattisgarh P.C.S. (Pre) 2018

Ans. (b)

The rotating armature of an AC generator is connected to an external circuit by the means of slip rings. In a DC generator, the commutator is used to reverse the directions of the natural alternating voltage every half cycle. So AC generator has slip rings while DC generator has a commutator.

74. Alternate current is not preferable –

- (a) To charge storage battery
- (b) To run/start a electric motor
- (c) To transmit electric power
- (d) To heat up electric toaster

R.A.S./R.T.S. (Pre) 1994

Ans. (a)

Direct Current is used to charging the batteries with the help of rectifier which converts AC to DC. Thus alternate current is not preferable for it.

75. The device used for converting alternating current to direct current is called –

- (a) Inverter
- (b) Rectifier
- (c) Transformer
- (d) Transmitter

Chhattisgarh P.C.S. (Pre) 2008

U.P.P.C.S. (Pre) 2006

Ans. (b)

Rectifier is an electric device that converts alternating current (AC) to direct current (DC).

Inverter is an electronic circuit that changes direct current (DC) to Alternating Current (AC).

The transformer supply alternating current or electric power from power grid to those appliances which use different voltages.

Transmitter is an electronic device which, with the aid of antenna produces electromagnetic signals/radio waves. Primarily it is used for radio, television and in other communication tools.

76. Alternating current is converted to direct current –

- (a) By Dynamo (b) By Motor
(c) By Transformer (d) By Rectifier

U.P. U.D.A./L.D.A. (Spl.) (Mains) 2010

Ans. (d)

See the explanation of above question.

77. Assertion (A): In India, electricity distribution companies calculate the consumption of electricity in kWh.

Reason (R) : Electrical system in India works at a frequency of 60 Hz.

Code :

- (a) both (A) and (R) are individually true, and (R) is the correct explanation of (A).
(b) both (A) and (R) are individually true, but (R) is not a correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

M.P.P.C.S. (Pre) 2008

Ans. (c)

50Hz frequency of power transmission is fixed in our country for normal operation of the power grid. In there an increase or decrease of 0.5 Hz is acceptable. But if it increases or decreases more than that limit, the grid gets failed. In India, electricity distribution companies calculate the consumption of electricity in kWh (units). Thus, (A) is true, but (R) is false.

78. A statement is followed by two arguments I and II. Choose which of the argument/s is/are strong :

Statement : Should all diesel engines be replaced by electric engines in trains?

Arguments :

I. Yes, diesel engine causes a lot of pollution.

II. No, India does not produce enough electricity to fulfil even the domestic needs.

- (a) Neither argument I nor argument II is strong.
(b) Both of the arguments are strong.

(c) Only argument II is strong.

(d) Only argument I is strong.

Rajasthan P.C.S. (Pre) 2023

Ans. (d)

All diesel engines should be replaced by electric engines in trains as diesel engine causes a lot of pollution. Thus, argument I is strong for the statement. While argument II is not strong for the statement.

79. Which of the following option represents the consumption of power in ascending order–

- (a) Television, fan, electronic press, electronic kettle
(b) Television, fan, electronic kettle, electronic press
(c) Fan, television, electronic press, electronic kettle
(d) Electronic press, electronic kettle, fan, television

U.P.P.C.S. (Pre) 1992

Ans. (c)

Manufacturer of electronic equipment sets a power rating for every equipment which shows the maximum power that can be consumed by the specific device. According to the question, the ascending or increasing order of the electronic items by their power rating are– fan, television, electronic press, electronic kettle.

80. The Earth's magnetic field is due to :

- (a) Currents circulating inside the core
(b) Presence of a huge magnet at its centre
(c) Moving charges in space outside
(d) None of the above

U.P.P.C.S. (Mains) 2011

Ans. (a)

The Earth's magnetic field is believed to be generated by electric currents in the conductive material of its core, created by convection currents due to heat escaping from the core.

81. What is the source of electrical energy in an artificial satellite ?

- (a) Solar cells (b) Mini nuclear reactor
(c) Dynamo (d) Thermopile

U.P.P.C.S. (Pre) 2014, 2006, 2000

M.P.P.C.S. (Pre) 2000

Ans. (a)

In artificial satellites solar (or photovoltaic) cells convert the Sun's energy into electrical energy through photovoltaic effect. The photovoltaic effect is the creation of voltage and electric current in a material upon exposure to light.

82. Consider the following statements :

Assertion (A) : Space-based solar power (SBSP), it is suggested be made a national goal.

Reason (R) : Supply of SBSP is 99% uninterrupted throughout the year, besides the enormity of energy availability.

Select the correct answer using the codes given below :

Code :

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

U.P.P.C.S. (Pre) 2010

Ans. (a)

Space-based solar power (SBSP) is a system for the collection of solar power in space for use on Earth. SBSP would differ from current solar collection methods in such a way that the means used to collect energy would reside on an orbiting satellite instead of the Earth's surface. The power is then transmitted via electromagnetic waves at 2.45 GHz to dedicated receiver stations on Earth. "Rectenna" convert the electromagnetic energy into electricity which is used in the local grid. The supply of SBSP will be 99% uninterrupted throughout the year besides the enormity. Former president Dr. APJ Abdul Kalam suggested that the space-based solar power (SBSP) should be made a national goal.

83. With reference to technologies for solar power production, consider the following statements :

1. 'Photovoltaics' is a technology that generates electricity by direct conversion of light into electricity, while 'Solar Thermal' is a technology that utilizes the Sun's rays to generate heat which is further used in electricity generation process.
2. Photovoltaics generates Alternating Current (AC), while Solar Thermal generates Direct Current (DC).
3. India has manufacturing base for Solar Thermal Technology, but not for Photovoltaics.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 and 3 only
- (c) 1, 2 and 3
- (d) None

I.A.S. (Pre) 2014

Ans. (a)

Photovoltaics refers to a technology which converts light into electricity using semiconducting materials that exhibit photovoltaic effect. Photovoltaic (solar) cells produce direct current electricity from sunlight. On the other hand, Solar Thermal Technology works on the principle of converting solar energy into heat energy by absorbing it which can be further used for electricity generation.

In March, 2013 Ministry of New and Renewable Energy, Government of India informed that India has capacity of generating 2000 MW electricity through solar photovoltaic modules. This clarifies that India has infrastructure for photovoltaic technology.

84. With reference to solar water pumps, consider the following statements:

1. Solar power can be used for running surface pumps and not for submersible pumps.
2. Solar power can be used for running centrifugal pumps and not the ones with piston.

Which of the statement given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

I.A.S. (Pre) 2020

Ans. (d)

Solar-powered pumps run on electricity generated by photovoltaic panels or the radiated thermal energy available from collected sunlight as opposed to grid electricity or diesel run water pumps. Solar power can be used for running surface pumps as well as for submersible pumps. Solar-powered submersible pumps support varied applications like field irrigation or supply potable water, while drawing water from underground water sources like borewells or sumps. Some solar submersible pumps use the same centrifugal mechanism as a surface pump, while others use a positive displacement mechanism. A solar-powered reciprocating pump is a class of positive displacement pump with piston, which is used for various purposes such as, car washing, irrigation, colour spraying, large fountain, garden water pump etc. From the above explanation it is clear that both of the given statements are incorrect.

85. Which one of the following elements is used in solar cells?

- (a) Silicon
- (b) Cerium
- (c) Astatine
- (d) Vanadium

Uttarakhand P.C.S. (Pre) 2012

Ans. (a)

Solar cells are the components of solar arrays that convert radiant light from the sun into electricity that is then used to power electrical devices. Solar cells contain materials with semiconducting properties in which their electrons become excited and turned into an electrical current when struck by sunlight. While there are a dozen of variation of solar cells, the two most common types are those made of crystalline silicon (both monocrystalline and polycrystalline) and those made with what is called thin film technology. The majority (90 %) of the solar cells on the market today are made of silicon.

86. Photovoltaic cells are :

- (a) Solar cells
- (b) Thermal cells
- (c) Sulphur cells
- (d) Molar cells

U.P. P.C.S. (Pre) 2016

Ans. (a)

The conversion of light (solar) energy into electrical energy is based on the phenomenon called photovoltaic effect. A photovoltaic cell is the basic unit of the system where the photovoltaic effect is utilized to produce electricity from light energy. Silicon is most widely used semiconductor material for constructing photovoltaic cell.

87. Photovoltaic cells is related to :

- (a) Nuclear Energy
- (b) Geo Thermal Energy
- (c) Solar Energy
- (d) Wind Energy

R.A.S./R.T.S. (Pre) 2016

Ans. (c)

See the explanation of above question.

88. Solar energy is converted into electric energy through:

- (a) Leclanche cells
- (b) Photovoltaic cells
- (c) Dry cells
- (d) Voltaic cells

U.P. P.C.S. (Pre) 2022

Ans. (b)

See the explanation of above question.

89. A photoelectric cell is a device which :

- (a) converts light energy into electric energy
- (b) converts electric energy into light energy
- (c) stores light energy
- (d) None of the above

69th B.P.S.C. (Pre) 2023

Ans. (a)

A photoelectric cell converts light energy into electric energy. The photoelectric effect, where electricity is generated from a material when it absorbs light, was first discovered in 1887 by Heinrich Hertz. The photovoltaic effect is closely related to the photoelectric effect, with a critical difference. In the photoelectric effect, electrons are emitted into space. But, in the photovoltaic effect, electrons enter what we call the conduction band of the material. The photoelectric device can capture high-energy photons and generate a high-voltage output, while the photovoltaic cell can capture low-energy photons and generate a high-current output. This combination can increase the power output and efficiency of solar PV systems.

90. Which of the following photoelectric devices is most suitable for digital applications?

- (a) Photodiode
- (b) Photovoltaic cell
- (c) Photoemitter
- (d) More than one of the above
- (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (b)

The photoelectric effect is the emission of electrons when electromagnetic radiation, such as light, hits a material. And photoelectric device means which uses photoelectric effect, in other words which produces electricity/electric signal from light. Photovoltaic cell is one such device. Photovoltaic cells or commonly known as solar cells have vast use in digital technology. For example, entire satellite communication depends on it, since this is the main source of power for it. Photoemitting or light emitting diodes emit light on flow of electricity i.e. reverse of photoelectric effect. Similarly, photodiode changes its electrical properties on receiving light. It is like light controlled switch which allows flow of electricity on getting light but it does not generate electricity. Hence, option (b) is the correct answer.

91. 'Net Metering' is sometimes seen in the news in the context of promoting :

- (a) the installation of CNG kits in motorcars
- (b) the installation of water meters in urban households
- (c) a billing mechanism for solar energy by consumers for the electricity they add to the grid
- (d) the use of piped natural gas in the kitchens of households

69th B.P.S.C. (Pre) 2023

Ans. (c)

'Net metering' is a billing mechanism that credits solar energy system owners for the electricity they add to the grid. For example, if a residential customer has a PV system on their roof, it may generate more electricity than the home uses during daylight hours. If the home is net-metered, the electricity meter will run backwards to provide a credit against what electricity is consumed at night or other periods when the home's electricity use exceeds the system's output. Customers are only billed for their 'net' energy use.

92. With reference to solar power production in India, consider the following statements :

1. India is the third largest in the world in the manufacture of silicon wafers used in photovoltaic units.

2. The solar power tariffs are determined by the Solar Energy Corporation of India.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2018

Ans. (d)

As per SEMI's World Fab Forecast Report, 2024, China led the world in global silicon wafer fab capacity installed, while Taiwan and South Korea ranks second and third, respectively. India is not among the leading manufacturers of silicon wafers in the world.

The solar power tariffs in India are determined by the Central Electricity Regulatory Commission (CERC) and State Electricity Regulatory Commissions (SERCs) on the basis of competitive bidding. Solar Energy Corporation of India (SECI) has a Power trading license, but it does not determine solar power tariffs.

93. Consider the following:

1. Battery storage
2. Biomass generators
3. Fuel cells
4. Rooftop solar photovoltaic units

How many of the above are considered "Distributed Energy Resources"?

- (a) Only one (b) Only two
(c) Only three (d) All four

I.A.S. (Pre) 2024

Ans. (d)

Distributed energy resources (DERs) are small-scale energy resources usually situated near sites of electricity use. Their rapid expansion is transforming not only the way electricity is generated, but also how it is traded, delivered and consumed. Examples of distributed energy resources that can be installed includes :

- rooftop solar photovoltaic units
- wind power generating units
- battery storage
- biomass generators, which are fuelled with waste gas or industrial and agricultural by-products
- open and closed cycle gas turbines
- reciprocating engines (diesel, oil)
- hydro and mini-hydro schemes
- fuel cells etc.

Hence, option (d) is the correct answer.

94. Recently, the term "pumped-storage hydropower" is actually and appropriately discussed in the context of which one of the following?

- (a) Irrigation of terraced crop fields
(b) Lift irrigation of cereal crops
(c) Long duration energy storage
(d) Rainwater harvesting system

I.A.S. (Pre) 2024

Ans. (c)

Pumped-storage hydropower (PSH) is a type of hydroelectric energy storage that uses water stored in two reservoirs at different elevations to generate electricity. When there is excess electricity available, such as during off-peak hours or from renewable sources like solar and wind, it is used to pump water from the lower reservoir to the upper reservoir. When there is a demand for electricity, the water is released from the upper reservoir back down to the lower reservoir, passing through turbines that generate electricity. PSH acts similarly to a giant battery, because it can store power and then release it when needed. Recently, the Ministry of Environment, Forest, and Climate Change (MoEFCC) has given the environment-related go-ahead to pumped storage hydropower projects with a capacity of 11.98 gigawatts (Gw) including Greenko Energy's proposal for a 3.66 Gw project in Uttar Pradesh's Sonbhadra district.

95. Which of the following statements is not true?

- (a) On the basis of wind power installed capacity, India ranks fourth in the world
(b) The total wind power installed capacity of India is about 400 GW
(c) Among Indian States, maximum wind power potential (at 100 m above ground level) is in Gujarat
(d) None of the above

Chhattisgarh P.C.S. (Pre) 2022

Ans. (b)

As on 28 February, 2025, the total wind power installed capacity of India is about 48.59 GW. Hence, statement of option (b) is not true. On the basis of wind power installed capacity, India ranks fourth in the world after China, USA and Germany in 2023. Among Indian States, maximum wind power potential (at 100 m above ground level) is in Gujarat.

96. Which of the following statements is not correct regarding wind power capacity in India?

- (a) Wind power capacity in India has significantly increased.
(b) Gujarat has around 22% of the total wind power capacity of the country.